REVIEW

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Continuous Downstream Processing for High Value Biological Products: A Review

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ABSTRACT: There is growing interest in the possibility of developing truly continuous processes for the large-scale production of high value biological products. Continuous processing has the potential to provide significant reductions in cost and facility size while improving product quality and facilitating the design of flexible multi-product manufacturing facilities. This paper reviews the current state-of-the-art in separations technology suitable for continuous downstream bioprocessing, focusing on unit operations that would be most appropriate for the production of secreted proteins like monoclonal antibodies. This includes cell separation/recycle from the perfusion bioreactor, initial product recovery (capture), product purification (polishing), and formulation. Of particular importance are the available options, and alternatives, for continuous chromatographic separations. Although there are still significant challenges in developing integrated continuous bioprocesses, recent technological advances have provided process developers with a number of attractive options for development of truly continuous bioprocessing operations.

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Introduction

The current paradigm for commercial-scale manufacture of high value biological products is based on batch processing in which each unit operation is completed in sequence, with the product outflow from one unit typically collected in a large holding tank before moving to the next step. This type of batch operation facilitates the design and optimization of the individual unit operations, and it easily accommodates off-line measurements of key product quality attributes before subsequent processing steps.

Recent years have seen growing interest in the possibility of operating specific steps in a continuous mode, with the long-term goal of moving towards an entirely continuous process in which the product moves directly from one step to the next without any significant intervening product hold. Ideally, the continuous process would operate at true steady-state, with the operating conditions (e.g., flow rates and pressures) and product/impurity concentrations remaining constant throughout the process. Continuous operation can also be achieved using periodic (cyclic) processes. The transition from batch to continuous processing revolutionized manufacturing in the chemical and pharmaceutical industries (La Porte, 2007). In some cases, this transition was highly disruptive. Many of the large US steel companies had made major investments in capital equipment for batch processing and were simply unable (or unwilling) to adapt and were ultimately outcompeted by companies employing continuous processes.

Although the details of each industry are clearly unique, the benefits of moving to continuous processes are similar and include:

- significant reductions in capital equipment costs and facility size,
- significant increases in productivity (largely because all equipment is in use at all times),
- greater flexibility (because of more efficient utilization of the facility),
- improved product quality (because of greater uniformity in process time).

The improvement in product quality could be particularly significant for complex biological products. For example, several studies have shown that protein aggregation and denaturation can occur when proteins are bound to chromatographic resins for long times due to protein unfolding and interactions with other proteins on the resin surface (Gillespie et al., 2012; Gospodarek et al., 2014; Guo and Carta, 2014). Batch operation of a packed chromatography column can thus generate a wide variation in product quality; the proteins that are initially loaded on the column remain in the bound state for well more than an hour, while proteins near the end of the load remain bound for only a small fraction of that time. Similar issues can occur during batch or fed-batch cell culture and subsequent product hold steps, leading to variability in protein glycosylation profiles (Pacis et al., 2011), extent of deamidation (Joshi et al., 2014), etc.

The technology for continuous cell culture was developed more than 30 years ago (Arathoon and Birch, 1986). The use of perfusion

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bioreactors with continuous addition of nutrients and removal of product can increase cell densities by as much as 10-fold, significantly increasing what were at the time very low product titers (typically $\leq$100 mg/L). The application of perfusion systems for monoclonal antibody (mAb) production is well-established (Konstantinov et al., 2006; Ozturk, 2014; Vogel et al., 2012; Warnock and Al-Rubeai, 2006).

This Review will focus on the state-of-the-art in downstream separation technologies that would be needed for continuous manufacturing of high value biological products. This discussion is organized around the critical steps in the downstream process: clarification (and cell recycle), initial product capture, product polishing, and formulation, with a focus on the processing of secreted products like monoclonal antibodies (Fig. 1). Jungbauer (2013) and Rathore et al. (2015) have considered the challenges in adapting many of the key process operations required for production of bacterial products in a continuous operation, including cell lysis and protein re-folding. The final section provides a brief review of issues related to virus clearance, which is critical for regulatory compliance for any biotherapeutic products.

Cell Recycle/Product Clarification

In principle, any of the processes used for product clarification could potentially be used for the initial separation in a continuous process. However, the need to recycle the retained cells places significant constraints on the operation since cell viability is critical to the long-term process. The net result is that centrifugation and depth filtration, which tend to dominate initial clarification in most batch mAb processes, are much less attractive in continuous applications. Voissard et al. (2003) provide a nice review of the available methods for cell retention in perfusion culture of mammalian cells, including centrifugation and gravity settlers (both vertical and inclined). The following discussion focuses on filtration and acoustic methods.

Spin Filters

Spin-filters are placed directly in the stirred tank reactor, typically mounted as an annulus around the impeller shaft at the center of the bioreactor, enabling continuous removal of the cell-free culture harvest from inside the rotating filter. Spin-filter technology was first introduced by Himmelfarb et al. (1969) using a stainless steel mesh covered by a membrane with 3 μm pore size. The filter was rotated by a magnetic stirrer at approximately 300 rpm. A number of studies have examined the effects of membrane pore size, filter geometry, and rotation rate on cell retention and fouling (Deo et al., 1996). Recent computational fluid dynamics (CFD) analyses have demonstrated the presence of a radial exchange flow in the spin-filter, with the overall hydrodynamics governed by the formation of Taylor–Couette vortices (Figueredo-Cardero et al., 2009). Although spin filters work well in small bioreactors, scale-up can be challenging since it is difficult to provide sufficient filter area to handle the volumes in large-scale bioreactors.

Femmer et al. (2015) developed an in situ membrane stirrer as an alternative to the classical spin-filter. In this case, a micromesh membrane is incorporated directly into the impeller blades used to provide agitation to the bioreactor (Fig. 2). The system is designed to provide aeration through a separate internal capillary; this not only provides the necessary gas transfer to the cells, but also reduces fouling and cell deposition on the membrane surface. Clarified product is withdrawn through the micromesh membrane into the impeller shaft and then out the top of the bioreactor. It is possible to backwash the membrane by reversing the flow direction, further extending the filter life. Excellent performance was obtained with a yeast cell suspension—future efforts will be required to demonstrate the effectiveness of this technology with mammalian cell lines used for recombinant protein production.

Submerged Membrane Filtration

Submerged membrane filtration recovers clarified cell culture fluid from the bioreactor using membranes that are placed directly in the bioreactor. A variety of geometries have been explored, including cylindrical rods and suspended hollow fibers (Carstensen et al., 2012). Hollow fiber systems have been adapted from the successful use of this technology in submerged membrane reactors for activated sludge (Meng et al., 2012). These narrow fibers have the size-selective skin on the outer surface of the fiber, with fluid withdrawn and collected through the fiber lumens (with one end of

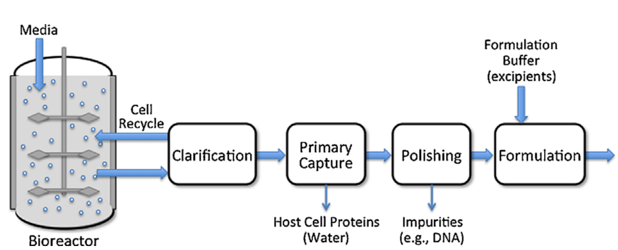

Figure 1. Schematic representation of a generic continuous downstream bioprocess.

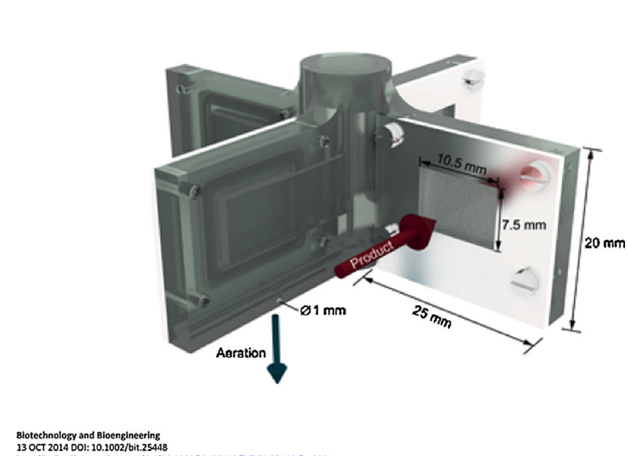

Figure 2. Integrated membrane stirred (reproduced from Femmer et al., 2015).

the fiber lumen closed by the potting). These systems have very high surface area per unit volume, allowing operation at low filtrate flux. The membranes can also be periodically back-flushed (flow from lumen to shell) to remove foulants from the membrane surface since the hollow fibers are self-supporting. In addition, aeration can be used to effectively scour the external membrane surface; the fiber motion induced by both aeration and agitation can further reduce cell deposition/adhesion to the membrane (Meng et al., 2012).

Tangential Flow Filtration

An alternative approach for initial clarification is to process some of the bioreactor contents in a separations unit that is outside of the bioreactor, often referred to as an "external loop" (Carstensen et al., 2012). The simplest method is to use conventional tangential flow filtration (TFF) modules, typically either large bore hollow fibers or open (non-screened) cassettes to minimize channel clogging (van Reis and Zydney, 2007). A small fraction of the bioreactor contents is continuously pumped through the TFF device; the retentate is directly returned to the bioreactor while the permeate (containing the secreted product) is taken for subsequent processing. The use of constant flux is most attractive for continuous processing, with the flux maintained by a second pump placed on the filtrate exit line. The TFF module is operated below the critical flux, that is, below the flux at which fouling first becomes measurable. This allows for stable long-term operation, although gradual fouling is still problematic. Considerable work is ongoing on the choice of membrane materials and pore size to reduce fouling and minimize product retention. The advantage of using an external filtration device is that the module can be readily replaced with minimal disruption to the bioreactor. However, continued recirculation of the bioreactor contents through the feed pump and TFF module can cause cell lysis, significantly complicating the operation of the perfusion bioreactor (Clincke et al., 2013).

Alternating Tangential Flow Filtration

Alternating tangential flow filtration (ATF) uses standard hollow fiber membranes to separate cells and product, but the flow is driven through the module using a very low shear diaphragm pump that cyclically withdraws and returns the cell suspension to the bioreactor (Clincke et al., 2013; Kelly et al., 2014). This eliminates cell damage due to passage through a peristaltic pump, and the alternating flow helps maintain the flux by "un-clogging" any blocked fibers. Typical cycle times are on the order of 1 min. Filtrate is withdrawn during both the pressure and exhaust cycles, typically using a roller pump connected to the filtrate exit (Fig. 3). The ATF has only a single connection to the bioreactor—cells and media are withdrawn and returned through the same port—easing operation and helping to maintain sterility. The hollow fiber module is easily replaced with minimal disruption to the bioreactor.

The ATF system has been operated continuously for weeks at a time, providing ideal integration with a perfusion bioreactor (Clincke et al., 2013). The reversing flow in ATF appears to minimize fouling, possibly due to Starling recirculation flow near the fiber exit, which could wash off deposited material. Kelly et al. (2014) showed that fouling during operation with a PER.C6 mammalian cell line was due to deposition of both cells and anti-foam micelles. More quantitative studies of fouling during long-term operation are needed to identify the key variables controlling the performance of the ATF system.

Non-Filtration Options

Although most applications of perfusion reactors use some type of size-based filtration for cell retention/product clarification, a variety of non-filtration options have also been explored. Wen et al. (2000) used a two-step sequential sedimentation system consisting of a traditional settling tank in series with a flat settler. Kilburn et al. (1989) used an acoustic system for cell separation, with the cells concentrated at the pressure nodes of the ultrasonic standing waves. This leads to coalescence of the cells into large aggregates that rapidly settle and facilitate their return to the bioreactor. Shirgaonkar et al. (2004) have provided a thorough review of acoustic cell separations. None of these systems has yet been applied in large-scale commercial bioprocessing applications.

Primary Capture

The dominant method for initial product capture is bind-and-elute chromatography, for example, the use of Protein A affinity resins for initial capture and purification of monoclonal antibodies. Protein A is particularly attractive since it requires no adjustment in pH or salt concentration prior to binding, it provides high levels of host cell protein and DNA removal, it is highly robust to small variations in feed and product characteristics, and it provides considerable volume reduction (Kelley, 2009). However, traditional packed column chromatography is inherently a batch process, with the column loaded, washed, and eluted in a fully sequential process (with the resin used for one operation at a time). As discussed below, a number of chromatographic methods have been developed to provide continuous product capture (although not always with true steady-state operation). In addition, the high titers that can be achieved with perfusion bioreactors using state-of-the-art cell lines and media could potentially enable the use of non-chromatographic product capture steps that are highly amenable to continuous processing.

Periodic Counter-Current Chromatography (PCC)

Periodic counter-current chromatography (PCC) was developed by GE Healthcare as a continuous bind-and-elute purification process (Bryntesson et al., 2011). PCC uses multiple packed columns operated in a series configuration with switching of the column feed, wash, and elution steps as shown schematically in Figure 4 for a three column process. In the first stage of operation, clarified cell culture fluid (CCF) is fed to Column 1 (with Columns 2 and 3 kept "idle"). As the first column approaches saturation, the outflow is loaded directly onto Column 2 to capture any product in the breakthrough from Column 1. When Column 1 is fully saturated, the CCF is loaded directly onto Column 2, with Column 1 washed using an appropriate wash buffer. The wash effluent from Column 1 is typically passed through Column 3 to capture any free product contained in the void space in Column 1. After washing, the product

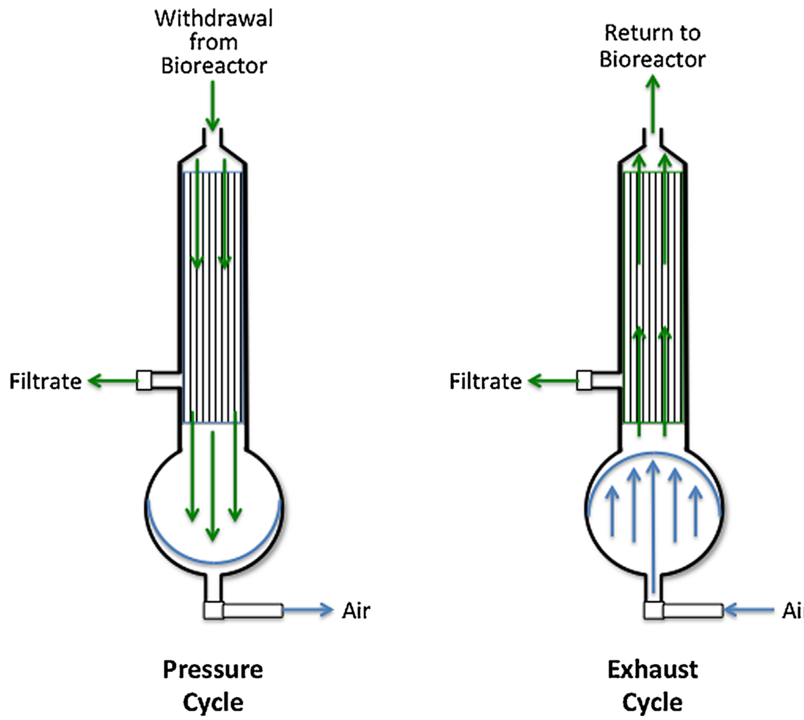

Figure 3. Pressure and exhaust cycles of an alternating tangential flow filtration (ATF) module.

is eluted from Column 1 using the desired elution buffer while Column 3 is brought on-line to capture any breakthrough from Column 2. Column 1 would then be regenerated and equilibrated, with the entire process repeated in a cyclical fashion to achieve continuous operation. Additional columns can be used to manage the different time/volume requirements for the binding, washing, elution, and regeneration steps.

The design of a PCC system must ensure that the time for protein loading (which could be distributed among more than one column) is greater than or equal to the time for recovery and regeneration (washing, elution, stripping, and equilibration). The minimum number of columns in a PCC system with a single loading column is thus (Godowat et al., 2012):

$$N \geq 1 + \frac{t_{\mathrm{recovery}}}{t_{\mathrm{R}}}\left(\frac{C}{q}\right) \tag{1}$$

where C is the product concentration in the feed, q is the binding capacity of the resin, t_{recovery} is the time needed for recovery and regeneration of a single column, and t_{R} is the required residence time for product binding (typically on the order of 2–8 min for most monoclonal antibodies). Note that processing high titer products using PCC requires more columns in the recovery and regeneration steps due to the reduction in loading time for the capture column.

Warikoo et al. (2012) demonstrated the integration of a perfusion bioreactor with a PCC system, with the column switching determined by on-line measurements of UV absorbance. Godowat et al. (2012) demonstrated the use of a 3-column periodic counter-current chromatography system for purification of a monoclonal antibody using Protein A (followed by iminodiacetate and hydrophobic interaction chromatography). PCC had higher resin capacity utilization and lower buffer consumption compared to conventional (single column) chromatography. Angarita et al. (2015) describe the use of CaptureSMB, a two-column system analogous to PCC, for initial capture of a mAb.

A number of other multi-column configurations have been explored, including the Simulated Moving Bed (SMB) Chromatography systems developed by Tarpon BioSystems (BioSMB$^{\mathrm{TM}}$, Worcester, MA), Semba Biosciences (Octave$^{\mathrm{TM}}$, Madison, WI), and Novasep (Shrewsbury, MA). SMB technology is discussed in the section on polishing since it is more commonly used in those operations. Although these multi-column systems can provide continuous separations for extended processing times, the overall process is periodic and does not provide true steady-state operation. In particular, the concentration of the eluted product varies cyclically throughout the process, decreasing from a high value at the start of the elution to a fairly dilute stream as the final product is eluted; this can be problematic for implementation of online process control and it can create challenges for subsequent processing operations (depending upon the details of the process). In addition, these systems have a very broad distribution in residence time for the bound product, which could potentially have a negative impact on product quality due to protein denaturation and/or aggregation (Gillespie et al., 2012; Gospodarek et al., 2014).

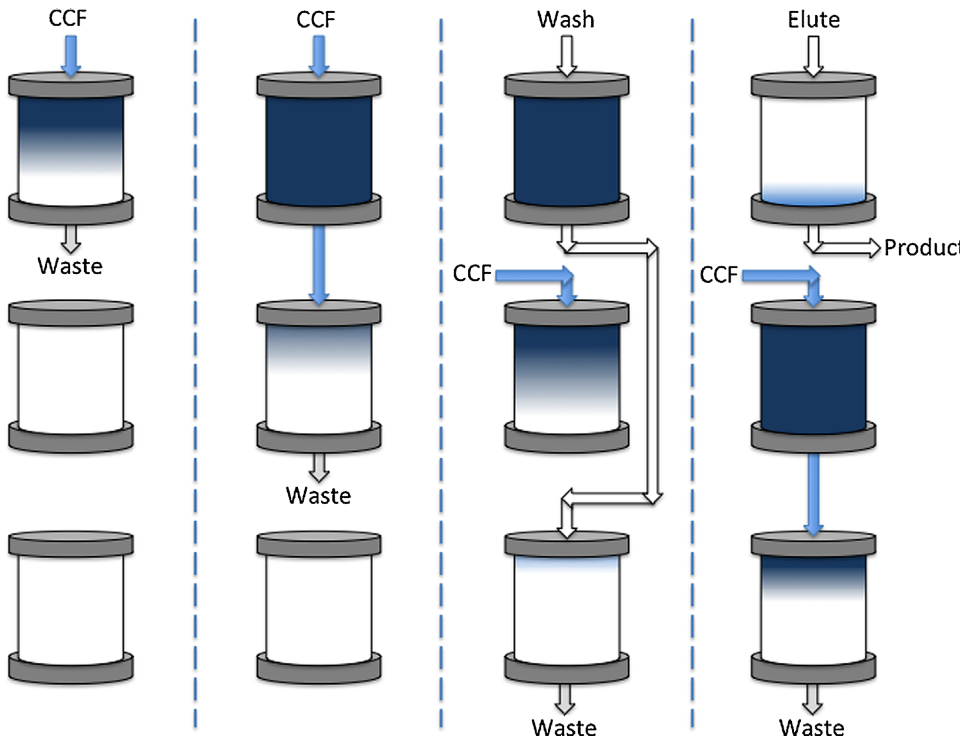

Figure 4. Cyclic operation of a 3-column periodic counter-current (PCC) column chromatography system.

Multicolumn Countercurrent Solvent Gradient Purification (MCSGP)

Multicolumn countercurrent solvent gradient purification (MCSGP) has similarities to both PCC and simulated moving bed (SMB) chromatography; it uses multiple columns in sequence that are switched in position to obtain countercurrent contacting (Aumann and Morbidelli, 2007). In contast to SMB, some of the columns in MCSGP are "short-circuited" to enable gradient elution; MCSGP can also be used for more than just binary separations. Muller-Spath et al. (2010a) have demonstrated the use of a 4-column MCSGP system for the initial capture of an IgG_2 mAb from clarified cell culture fluid using cation exchange resins with a gradient elution. Krättli et al. (2013) discussed the use on online control in the operation of a twin-column MCSGP system. MCSGP has yet to be implemented at commercial scale.

Continuous MCSGP can also be used for polishing operations, including those involving more complex separations/gradients. For example, Muller-Spath et al. (2010b) demonstrated the effectiveness of continuous MCSGP using cation exchange resins for the separation of charged variants of several commercial mAbs (Avastin, Herceptin, and Erbitux). ChromaCon (Zurich, Switzerland) has commercialized a 2-column MCSGP system called Contichrom® (Ulmer et al., 2015).

Continuous Countercurrent Tangential Chromatography (CCTC)

Continuous countercurrent tangential chromatography (CCTC) utilizes the chromatographic resin in the form of a slurry which flows sequentially through a series of static mixers and hollow fiber membrane modules (Napadensky et al., 2013; Shinkazh et al., 2011). The micro-porous hollow fiber membranes retain the large resin particles while letting all dissolved species, including proteins and buffer components, pass through the membrane and into the permeate. The buffers used in the binding, washing, elution, and regeneration steps flow countercurrent to the resin slurry in a multi-stage configuration within each step (Fig. 5), enabling high yield and purification while reducing buffer requirements. In contrast to the periodic behavior of multi-column chromatography systems, CCTC provides truly continuous steady-state operation. The product obtained from the elution step is at constant concentration and quality due to the uniformity in residence time in the binding, washing, and elution stages. This greatly facilitates integration with a perfusion bioreactor as shown in Figure 1.

Impurity removal and product yield are determined by the ratio of the permeate (q_p) to feed (q_F) flow rates, the conversion in the hollow fiber modules, and the number of stages in the washing and elution steps. For elution, the product yield is given as (Shinkazh et al., 2011):

$$Y = \frac{\alpha(\alpha^N - 1)}{\alpha^{N+1} - 1} \tag{2a}$$

with

$$\alpha = \frac{(q_p/q_F)}{1 - \varphi(1 - \epsilon)} \tag{2b}$$

where φ is the volume fraction of solids (resin) in the slurry and ε is the void volume in the resin particles. For $\alpha = 3$, the product yield

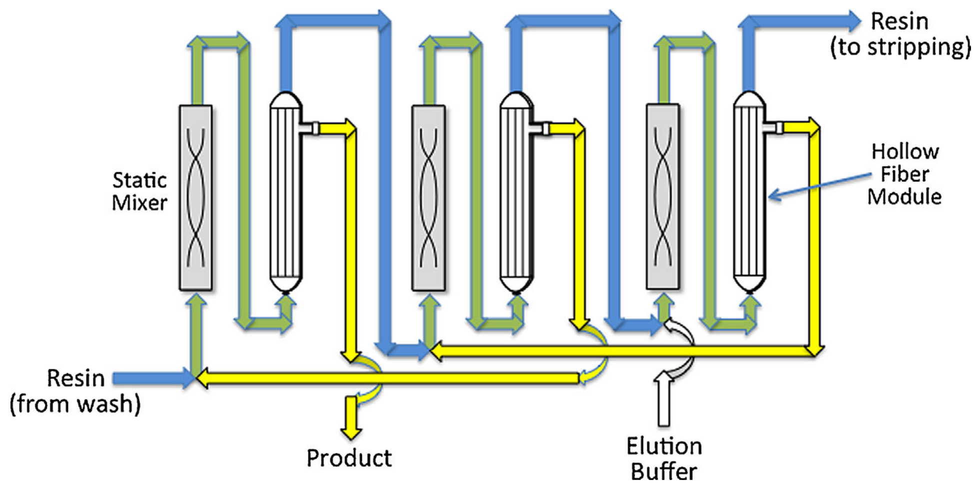

Figure 5. Three-stage elution step in continuous countercurrent tangential chromatography (CCTC).

increases from 75% for a 1-stage elution to 92% for 2 stages, and 98% for 3 stages. Increasing the number of stages can be used to reduce buffer requirements in the washing and elution steps.

Dutta et al. (2015) demonstrated the capability of using CCTC for initial capture and purification of two commercial mAbs from clarified cell culture fluid. Host cell protein removal, antibody yield, and product purity for both a low titer (~0.67 g/L) and high titer (~6.9 g/L) mAb were similar to that obtained with packed column Protein A chromatography, demonstrating the versatility of the CCTC system. The concentration of the eluted product remained constant (within ± 0.1 g/L) throughout the CCTC process, with operating pressures less than 35 kPa (5 psi). In addition, the productivity of the CCTC system, defined as the mass of purified mAb per gram of resin per hour, was several times greater than that obtained with a packed (batch) column.

Non-Chromatographic Processes

Although initial product capture is currently dominated by chromatographic separations, a number of alternative technologies could become attractive at the high product titers that can be achieved using optimized media and cell lines. For example, protein precipitation tends to become more selective as the product titer increases. Hammerschmidt et al. (2014) developed a continuous precipitation process for purification of a monoclonal antibody using calcium chloride and caprylic acid for precipitation of key contaminants followed by poly(ethylene glycol) and cold ethanol for precipitation of the mAb. Product yield and purity were similar to a platform mAb purification process using Protein A. Precipitation was accomplished in a continuous tubular reactor, consisting of a large tube with a static mixer insert to obtain uniform residence time, with the precipitate recovered by filtration. This provided uniform precipitate yields and morphologies, and it avoids the local denaturation that can sometimes occur with batch processes.

Aqueous two-phase systems (ATPS) can also be used in a continuous fashion by employing appropriate column contactors, spray columns, rotating disk contactors, or mixer-settler units (Espita-Saloma et al., 2014). Product separation in ATPS is based on differential partitioning between two aqueous phases, for example, the two phases formed from mixtures of immiscible polymers like polyethyleneglycol (PEG) and dextran or from phase-separating mixtures of PEG and appropriate salts (Goja et al., 2013). Eggersgluess et al. (2014) developed a continuous countercurrent multi-stage aqueous two-phase extraction process using a mixer-settler in combination with an extraction column. Good purification and concentration were obtained for the initial purification of a monoclonal antibody from clarified cell culture fluid.

Polishing

Polishing steps are typically designed to remove specific impurities, for example, DNA, endotoxin, soluble aggregates, and/or charge variants. It is also necessary to remove leached Protein A (assuming that Protein A is used for the initial capture). Chromatography remains the dominant method for product polishing, using either flow-through mode (impurities bound to the resin with the product collected in the flow-through) or differential chromatography, with the separation due to differences in the strength of interactions between the product/impurities and the solid phase. Several studies have examined the use of Multicolumn Countercurrent Solvent Gradient Purification (MCSGP) for polishing operations (Muller-Spath et al., 2008). There is also interest in using crystallization for final purification, although this technology has typically been most successful for small proteins like insulin. Recent work by Huettmann et al. (2014) has demonstrated the use of simultaneous crystallization and aqueous two-phase extraction for the purification of a single chain monoclonal antibody.

Flow-Through Chromatography

Although flow-through chromatography is generally performed as a batch process, it can easily be operated in what amounts to a continuous mode by having two columns in parallel, with the feed switched from one column to the other when the first column is nearly saturated with impurities. Flow-through chromatography can be performed using conventional bead-based media, although monolithic columns (Etzel and Riordan, 2009) and membrane adsorbers (Weaver et al., 2013) are particularly attractive since the lower binding capacities in these systems are generally much less of an issue for impurity removal and the flow-through mode leverages the high linear velocities that can be achieved due to the accessibility of the binding sites in convective flow-through pores. Application of membrane adsorbers in continuous processes would typically be done using constant flow operation, although membrane fouling can be problematic with some feeds. Membrane adsorbers are typically designed for single-use operation and would simply be discarded when saturated with impurities. Ion exchange media are most common, although hydrophobic interaction membranes are also available. Etzel and Riordan (2009) showed very high levels of DNA and virus removal using a monolith with a quaternary amine functionality.

Simulated Moving Bed (SMB) Chromatography

Simulated moving bed (SMB) chromatography can provide continuous (cyclic) operation for the separation of binary mixtures by differential chromatography, for example, for removal of product variants or oligomers (Aniceto et al., 2015). SMB systems use multiple packed columns with periodic switching of the feed, mobile phase, and recovery ports in a fashion that simulates countercurrent flow (but without any motion of the solid phase). A typical 4-column configuration is shown in Figure 6. At the start of the process, the feed (a binary mixture of product and impurity) is loaded to Column 1 in the upper right hand corner. Both components migrate through the column, with the more weakly bound species (in this case the product) recovered through the port at the column exit. Additional mobile phase is added between Columns 2 and 3, with the more strongly bound component recovered from Column 3. When the product has migrated much of the way through Column 1, the location of the feed, mobile phase, and recovery ports are rotated clockwise (in the same direction as the flow) to simulate the countercurrent movement of the solid phase. This type of continuous operation is only possible if the loading time is greater than the regeneration time. SMB operates cyclically, with the concentration of recovered product decaying from its maximum value at the start of each cycle to its minimum right before switching.

SMB chromatography has been successfully implemented in large-scale manufacturing for the separation of enantiomers (Rajendran et al., 2009), fructo-oligosaccharides (Vankova et al., 2008), and lysine (Van Walsem and Thompson, 1997), among others. Several SMB systems have also been developed for protein purification, including the Semba Octave Chromatography System and Tarpon Biosystems BioSMB. However, commercial applications have been limited due to concerns about system complexity and potential issues with the large number of switching valves. Zobel et al. (2014) have recently described a single-column SMB system with six containers and two pumps to collect and distribute the feed, product, and impurities.

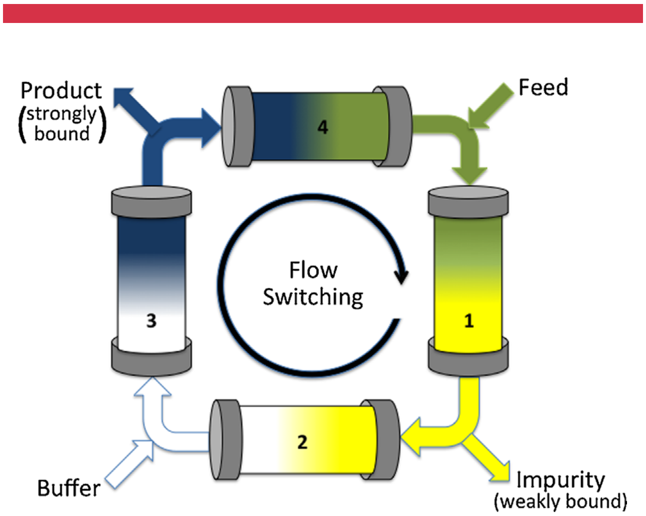

Figure 6. Schematic of 4-column simulated moving bed (SMB) chromatography system.

Annular Chromatography

In contrast to SMB, annular chromatography provides truly continuous purification using a rotating column (Uretschlager and Jungbauer, 2002). The feed is introduced at a fixed location in the annular space between two rotating concentric cylinders (Fig. 7). Product is recovered at fixed outlet ports at the bottom. Annular chromatography has been used for removal of protein aggregates from an intravenous (serum) immunoglobulin preparation by size exclusion (Buchacher et al., 2001) and for the continuous purification of recombinant Factor VIII produced in a perfusion bioreactor using a weak anion exchange resin (Vogel et al., 2002),

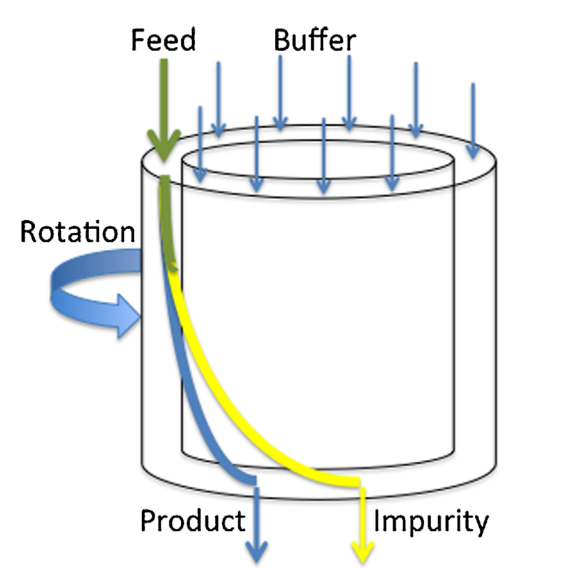

Figure 7. Schematic of rotating annular chromatography system.

among others. Obtaining uniform column packing and flow distribution remain challenging; the technology has not yet been implemented for large-scale protein purification.

Radial Flow Chromatography

Radial flow chromatography has many similarities to annular chromatography except that the flow in the annular bed is in the radial direction, moving from the outer radius to the inner collection area (Lay et al., 2006). Flow from the inner to outer cylinder is also possible. The feed, wash buffer, and elution buffer are introduced at different fixed angular positions, with the product (and impurities) collected at different angular positions (Lay et al., 2006). Continuous operation can be achieved by rotating the annulus past the fixed positions of the feed and buffer inlets and product collection ports (Lay et al., 2006). Radial flow chromatography systems have short bed heights, allowing operation at relatively low pressure drops. In addition, the radial configuration provides high cross-sectional area that can more easily accommodate high feed flow rates than conventional columns (Cabanne et al., 2007). Although a number of companies provide radial flow chromatography systems for batch operations, e.g., Sepragen (Hayward, CA) and Proxcys (Nieuw-Amsterdam, Netherlands), there do not appear to be any commercial-scale continuous (rotating) radial flow chromatography systems available for bioprocessing applications. It is also possible to use radial flow in membrane chromatography, with the membranes spiral wound or pleated around a central core (Ghosh et al., 2013).

Protein Crystallization

Protein crystallization can be used for both product purification and formulation; the regular lattice structure is able to exclude impurities and misfolded/aggregated product (Zang et al., 2011). Huettmann et al. (2014) demonstrated the use of crystallization for the purification of a single chain antibody obtained from an aqueous two-phase system, and Zang et al. (2011) explored conditions for purification of an IgG4 monoclonal antibody. However, the only biopharmaceutical currently produced using crystallization at industrial scale is insulin. One of the challenges in crystallization is the large range of variables that need to be optimized and controlled, including pH, temperature, buffer type and concentration, and excipients.

Although crystallization is often conducted in batch mode, continuous crystallization can provide enhanced product quality and better control of crystal morphology (Kwon et al., 2014). Continuous crystallization can be done using mixed suspension—mixed product removal reactors, consisting of a stirred tank with continuous removal of crystals. Although simple to operate, these systems have long residence time distributions leading to a distribution in crystal size and shape. Plug flow crystallizers using a tube with a static mixer can provide continuous crystallization with much more uniform product quality (Alvarez and Myerson, 2010).

High Performance Tangential Flow Filtration

High performance tangential flow filtration (HPTFF) can provide high resolution protein separations based on differences in both size and electrical charge of the product and impurity (Zydney and van Reis, 2001). The high selectivity in HPTFF is achieved by exploiting a number of phenomena. HPTFF operates in the pressure-dependent regime at or below the "transition point" in a plot of filtrate flux versus transmembrane pressure to minimize fouling and increase selectivity. In addition, the buffer pH and ionic strength are adjusted to maximize differences in effective volume of the product and impurities. Optimal performance is typically attained by operating close to the isoelectric point (pI) of the lower molecular weight protein and at relatively low salt concentrations (around 10 mM ionic strength); this maximizes electrostatic exclusion of the larger and more highly charged species. Direct charge–charge interactions can also be exploited by using a charged membrane to increase retention of all species with like-polarity (Zydney and van Reis, 2001).

HPTFF has been successfully used to separate monomers from oligomers, an antigen binding fragment from *E. coli* host cell proteins (Lebreton et al., 2008), a monoclonal antibody from mammalian cell host proteins (van Reis and Zydney, 2007), and a singly pegylated protein from more highly pegylated species (Ruanjaikaen and Zydney, 2011). Although HPTFF is currently performed as a batch operation, with the separation effected by diafiltration, it should also be possible to design a continuous HPTFF process using a membrane cascade analogous to that used previously for continuous membrane separation of oligosaccharides (Patil et al., 2014) and roxithromycin (Peeva et al., 2014). The design of a continuous membrane cascade is discussed in Cascade Diafiltration Section.

Formulation

Batch ultrafiltration/diafiltration is currently used for final formulation of nearly all recombinant protein products (van Reis and Zydney, 2007). This typically involves both product concentration (ultrafiltration) and buffer exchange (diafiltration). For monoclonal antibodies, the protein concentration in the final formulated product is typically on the order of 200 g/L; these high concentrations are needed to provide the desired therapeutic dosage in the limited volumes that can be delivered via subcutaneous injection (Shire et al., 2010). Several different approaches can be used to develop continuous ultrafiltration and diafiltration processes, including single pass tangential flow filtration systems and membrane cascades. In addition, it is possible to develop continuous versions of alternative methods for final product formulation, for example, lyophilization or crystallization.

Single-Pass Tangential Flow Filtration (SPTFF)

Single pass tangential flow filtration refers to any TFF system in which the conversion in a single pass through the module (permeate flow rate divided by inlet feed flow rate) is large enough that the system can be operated without a recycle (recirculation) loop for the retentate. SPTFF is thus directly applicable to continuous processes—the feed is simply pumped directly through the module with the concentrated product obtained from the retentate exit. SPTFF can be used for in-line concentration between other unit operations, for example, to reduce process volume prior to a chromatography or precipitation step, or it can be used to achieve the required product concentration in the final formulation. Note that the product in SPTFF only makes a single pass through the membrane module (and thus a single pump pass), which could reduce product aggregation/denaturation.

Although any membrane module could potentially be used in single pass mode, the high conversion of feed to permeate will cause a significant reduction in retentate flow rate along the length of the module. The Pall Cadence^{TM} system (Pall Corp., Port Washington, NY) consists of two or more internal stages with sequentially decreasing cross-sectional area; this compensates for the reduction in retentate flow rate enabling very high conversions at reasonable values of the filtrate flux. Casey et al. (2011) used a Pall Cadence^{TM} SPTFF module to concentrate serum IgG from 45 g/L to more than 200 g/L in a single pass with greater than 95% product recovery. Dizon-Maspat et al. (2012) examined the use of SPTFF for in-line concentration of three recombinant monoclonal antibodies to reduce process volumes from 2–10x between different chromatography steps.

Cascade Diafiltration

In order to achieve high levels of buffer exchange, that is, nearly complete removal of salts and buffer components from previous processing steps, most products are formulated using diafiltration with between 10 and 20 diavolumes (van Reis and Zydney, 2007). The concentration of a small solute (no retention) will be reduced by a factor of 10^4 after 9.2 diavolumes; a one-million fold reduction requires nearly 14 diavolumes. Diafiltration is an inherently batch process; the feed is continuously cycled through the membrane module while the formulation buffer is added at a rate equal to the permeate flow rate to maintain constant volume. Although buffer exchange could be accomplished using SPTFF by inline mixing of the formulation buffer with the feed, a 10^4 reduction in solute concentration would require a ratio of buffer to feed flow rates of 9,999, which is completely impractical. Even repeated application of SPTFF would be problematic. For example, four SPTFF modules, each with 90% conversion would be needed to achieve a 10^4 reduction in impurity levels.

Continuous diafiltration can be performed using a cascade of membrane modules arranged with countercurrent contacting (Peeva et al., 2014). In this case, the feed is continuously pumped into the final module in the cascade, with the retentate from each stage recycled to the previous stage in the cascade (Fig. 8). The diafiltration buffer is added to the first stage, with the diafiltered product removed in the first retentate. The fractional removal of a small impurity can be evaluated using the general equation for countercurrent staging as:

$$f = \frac{1-\beta}{1-\beta^{N+1}} \tag{3}$$

where β is the ratio of the diafiltration buffer flow rate to the feed flow rate. A 3 stage system with $\beta = 10$ would yield $f = 9 \times 10^{-4}$; this can be improved to $f = 9 \times 10^{-6}$ using 5 stages. The buffer requirement can be reduced, at the same level of impurity removal, by using more stages. Note that the membrane modules in this diafiltration cascade would need high conversion for effective impurity removal; a system with $\beta = 10$ would require a conversion of 91%.

Alternative Formulation Methods

Although ultrafiltration/diafiltration is currently the method of choice for formulation of essentially all therapeutic proteins, highly concentrated protein formulations can also be achieved by spray drying, fluidized bed drying, vacuum drying, lyophilization, or crystallization (Shire et al., 2010). Protein crystallization was discussed in Protein Crystallization Section. Lyophilization (or freeze-drying) is typically done in batch mode directly in vials or in vertical trays. Awotwe-Otoo et al. (2012) have discussed the use of quality-by-design strategies to optimize the performance of a batch lyophilization process for formulation of a monoclonal antibody product. Weisselberg (2013) has described a continuous lyophilization process in which the product solution cascades through a series of vertically stacked trays. This technique was demonstrated for lyophilization of food products; the current form of this technology would not meet the sterility requirements for production of biopharmaceuticals. GEA Niro (Columbia, MD) has developed a continuous lyophilization system (CONRAD) in which the product is simply moved through a long tubular lyophilization chamber on a tray. Although this system is currently only used for food products (e.g., freeze-dried coffee), it might be adaptable to continuous bioprocessing with appropriate modifications to insure product sterility. Rey (2010) has discussed the possibility of using similar systems for lyophilization of biological products, although this has not yet been demonstrated in practice.

Viral Clearance

Viral clearance is a critical component of downstream processing for injectable products. Current regulatory guidelines require at least two orthogonal steps, each of which deliver at least 4-logs of virus removal, that is, a log-reduction value or LRV of at least 4. Most mAb processes use a low pH hold for virus inactivation; this is particularly convenient in antibody processing since the mAb is eluted from the Protein A column at low pH. Significant viral clearance can also be achieved in chromatographic processes (Miesegaes et al., 2010), although most manufacturers also employ a virus filtration step to provide a robust, primarily size-based,

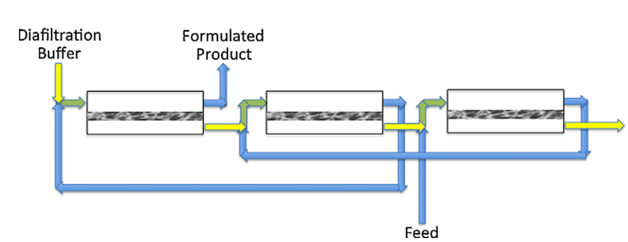

Figure 8. Three-stage countercurrent arrangement for continuous diafiltration.

method for virus removal. Although virus filtration cannot be directly converted to continuous operation due to issues of membrane fouling and LRV decline (Jackson et al., 2014), it would be possible to operate two virus filters in parallel with the feed switched to the second filter when the capacity of the first filter was achieved.

Viral inactivation is currently performed as a batch process, but it would be relatively straightforward to develop continuous inactivation processes using a tubular "reactor" with a static mixer to achieve uniform residence time. Low pH hold is typically done for 60 min, which would require a "reactor" volume of 0.4 L for a continuous process operating at 10 L/day. Virus inactivation by UV irradiation is dependent on the dose (in mJ/cm^2). Continuous flow UV reactors for virus inactivation are already well-established in water treatment applications (Timchak and Gitis, 2012). Li et al. (2005) have discussed the application of continuous flow UV for inactivation of a range of mammalian viruses (CPV, hepatitis A, pseudorabies virus, and bovine viral diarrhea virus). Bayer Technology Services (Leverkusen, Germany) has developed a helical UV reactor specifically for bioprocessing applications in which the UV source is located along the core of the helix, with mixing provided by induced Dean vortices (Bae et al., 2009).

Conclusions

Although continuous processes have yet to be employed for large-scale purification of therapeutic proteins, recent advances in downstream processing technology have created significant opportunities for the realization of fully integrated continuous processes. Warikoo et al. (2012) recently presented results for the capture and initial purification of a monoclonal antibody and a recombinant human enzyme produced using a perfusion bioreactor operated with an alternating tangential flow filtration (ATF) system and directly integrated with a periodic counter-current chromatography (PCC) system. The integrated processes ran continuously for 30 days with comparable product quality to traditional batch processes but with much smaller equipment footprint and greater productivity. Future efforts will be needed to determine which of the downstream processing options discussed in this Review can be most effectively employed in the design and operation of continuous processing for commercial scale manufacturing of high value biological products.

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